

Cisco Codec EQ: Cross View and Presenter Track Camera — xAPI Reference

Document version: 1.0 | Applicable devices: Cisco Codec EQ (RoomOS) | Reference: [RoomOS xAPI](#)

1. Overview

The **Cisco Codec EQ** is a RoomOS-based video conferencing endpoint that supports:

- **Cross View** — AI-driven cinematic meeting views using multiple cameras (PTZ 4K or Quad).
- **Presenter Track** — Camera follows the active presenter when using a PTZ 4K or Precision 60 camera.

Control is via **xAPI** (RoomOS XML/HTTP API): Commands, Configuration, Status, and Events.

2. Cross View

2.1 What Is Cross View?

Cross View uses multiple cameras and room analytics to analyze room activity, recognize and track participants, and select optimal camera angles.

2.2 Typical Codec EQ Cross View Setup

Component	Role
Codec EQ	Main room codec
Two side cameras	PTZ 4K or Quad (left/right)
One Quad camera	Above the display
Ceiling or Table Microphones Pro	Speaker location and framing

2.3 Setup and Calibration

- Follow Cisco **Multi Camera Setup Guidelines**.
- Use the **Multi camera director** setup wizard.
- Use the calibration board during the wizard.
- Configure SpeakerTrack with external mics (see SpeakerTrack xAPI below).

2.4 Related xAPI for Cross View

- Cameras — roles, placement, presets

- Cameras SpeakerTrack — tracking, frames, meeting zone
- Video Input — main video source
- `Configuration.Cameras.Camera[n].Role`

3. Presenter Track

3.1 What Is Presenter Track?

Uses a **PTZ 4K** or **Precision 60** to detect a person in the presenter zone, track the head, and keep them framed. Turns off when changing presets/camera/standby (unless Persistent).

3.2 Enable Presenter Track (Configuration)

xAPI Path	Description
<code>Configuration.Cameras.PresenterTrack.Enabled</code>	Enable/disable
<code>Configuration.Cameras.PresenterTrack.Connector</code>	PTZ/presenter camera connector
<code>Configuration.Cameras.PresenterTrack.TriggerZone</code>	Trigger zone for detection
<code>Configuration.Cameras.PresenterTrack.CameraPosition.Pan/Tilt/Zoom</code>	Default/limit positions

3.3 Command.Cameras.PresenterTrack.Set

Set presenter tracking mode (PTZ 4K or Precision 60 required; Presenter Track must be enabled).

Syntax (JavaScript):

```
xapi.Command.Cameras.PresenterTrack.Set({ Mode: value });
```

Modes:

Mode	Description
Off	Turn off presenter tracking
Follow	Turn on tracking (default from UI)
Diagnostic	Show trigger zone and tracking on screen for troubleshooting
Background	Tracking on without cropping (internal scripts)
Setup	Show trigger zone during setup (web UI)
Persistent	Tracking stays on until preset change etc.

Examples:

```
xapi.Command.Cameras.PresenterTrack.Set({ Mode: 'Follow' });
xapi.Command.Cameras.PresenterTrack.Set({ Mode: 'Off' });
xapi.Command.Cameras.PresenterTrack.Set({ Mode: 'Diagnostic' });
```

3.4 Other Presenter Track Commands

- `Command.Cameras.PresenterTrack.StorePosition()` — Store current camera position
- `Command.Cameras.PresenterTrack.ClearPosition()` — Clear stored position

3.5 Presenter Track Status

xAPI Path	Description
<code>Status.Cameras.PresenterTrack.Availability</code>	Feature available
<code>Status.Cameras.PresenterTrack.Status</code>	Current status
<code>Status.Cameras.PresenterTrack.PresenterDetected</code>	Presenter currently detected

4. SpeakerTrack (Multi-Camera / Cross View)

SpeakerTrack controls speaker tracking and view composition; used with Cross View.

Commands (selected): `Command.Cameras.SpeakerTrack.Activate` , `Deactivate` , `Set` , `Closeup.Activate/Deactivate` , `Frames.Activate/Deactivate` , `ViewLimits.Activate/Deactivate` , `ViewLimits.StorePosition` , `Whiteboard.StorePosition` , `Whiteboard.ActivatePosition` .

Configuration (selected): `Configuration.Cameras.SpeakerTrack.Mode` , `Closeup` , `DefaultBehavior` , `TrackingMode` , `Frames.Mode` , `Frames.RowConfiguration` , `MeetingZone.Mode/Area` , `MultiCamera.CalibrationMonitoring` .

Status: `Status.Cameras.SpeakerTrack.Availability` , `State` , `Status` , `ActiveConnector` , `Frames.Availability` , `Frames.Status` , `ViewLimits.Pan/Tilt/Zoom` .

5. Camera Configuration (Codec EQ)

Configuration: `Configuration.Cameras.Camera[n].Role` , `PtzAutoFraming` , `DefaultPosition.Pan/Tilt/Zoom` , `Placement.X/Y/Z` , `Focus.Mode` , `Standby.Mode` .

Status: `Status.Cameras.Camera[n].Connected` , `Position.Pan/Tilt/Zoom` , `Model` , `DetectedConnector` .

6. Connecting to xAPI

SSH, HTTP/HTTPS (e.g. POST to `/putxml`), WebSocket, or RS-232.

Example HTTP (XML):

```
POST /putxml HTTP/1.1
Host: <codec-ip>
Content-Type: application/xml

<Command><Cameras><PresenterTrack><Set><Mode>Follow</Mode></Set></PresenterTrack></Cameras></Command>
```

7. Quick Reference — Presenter Track

Goal	xAPI
Turn on	<code>Command.Cameras.PresenterTrack.Set({ Mode: 'Follow' })</code>
Turn off	<code>Command.Cameras.PresenterTrack.Set({ Mode: 'Off' })</code>
Troubleshoot	<code>Command.Cameras.PresenterTrack.Set({ Mode: 'Diagnostic' })</code>
Enable feature	<code>Configuration.Cameras.PresenterTrack.Enabled = true</code>
Presenter detected?	<code>Status.Cameras.PresenterTrack.PresenterDetected</code>
Store position	<code>Command.Cameras.PresenterTrack.StorePosition()</code>

8. References

- RoomOS xAPI: roomos.cisco.com/xapi
- PresenterTrack.Set: roomos.cisco.com/xapi/Command.Cameras.PresenterTrack.Set/
- Cameras domain: [roomos.cisco.com/xapi \(Cameras\)](https://roomos.cisco.com/xapi (Cameras))
- Cisco Multi Camera Setup / Cross View guidelines (help.webex.com, cisco.com)
- RoomOS API Reference Guide (PDF) — Cisco documentation portal

This document is a concise reference for Cisco Codec EQ Cross View and Presenter Track with xAPI. For exact parameters and product support, use the official RoomOS xAPI reference.